A Statistical Analysis of Wildland and Prescribed Fires in the Changing California Climate

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With climate change-driven large, destructive wildfires becoming more frequent, especially in California, using data to identify effective or ineffective prevention strategies, such as prescribed burns, can inform future prevention strategies. There exists a large body of research demonstrating the effectiveness of prescribed burns for mitigating the effects of large wildfires; however, much of this research was performed in the previous century and did for climate change's impact on prescribed burn efficacy. Additionally, the data needed for such an analysis is fragmented and complex, requiring the synthesis of wildfire, prescribed fire, and climate data from various sources. Statistical data science methods provide tools for making sense of the immense heterogenous data relevant to analyzing the relationship between prescribed fires, wildfires, and the climate.

In this work, we present an initial statistical analysis on a new dataset created from fusing prescribed fire and wildland fire records from The California Department of Forestry and Fire Protection (CalFire) [1, 2] with climate data from the National Oceanic and Atmospheric Administration (NOAA) [3]. A suite of statistical methods, including descriptive, inferential, and predictive methods are applied to the dataset to analyze if prescribed burns are effective in mitigating large wildfires in recent history while accounting for climate change. Descriptive methods are used to summarize the data trends, while inferential methods identify significant trends. Inferential methods include non-parametric tests, including the Mann-Whitney U test for differences in acres burned between predictor levels, and parametric tests, such as the χ^2 test for independence of region on number of fires. A time series analysis with cross correlation is used to identify which variables are related to wildfire acres burned at different time lag values. Additionally, a regression analysis is performed to identify significant predictors of wildfire acres burned, as well as to estimate expected acres burned when climate variables, such as average temperature and precipitation, change in accordance with climate models. When grouped across the entire state with a monthly time scale, results indicate prescribed burns do not impact the number of acres burned due to wildfire. However, when considering smaller regions, the impact of prescribed burns is measurable. At a state-wide level, the Palmer Drought Severity Index (PDSI), average temperature, maximum temperature, and precipitation are related to wildland fire acres burned, with increased temperature, increased drought, and lower precipitation positively correlated with wildfire acres burned. Overall climate variables have a stronger impact on wildfire size, though prescribed burns are associated with slightly smaller wildfires.

References

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